ELMROW

Encryption compared with Signal

October 24, 2026. Read from the Elmrow repo, not from a product claim.

Elmrow is not the Signal Protocol. A fresh pass through the sealing code changes one earlier claim: the first letter is stricter than a short summary suggested, and the later letters are weaker. The server cannot read a letter. That is not the same as matching Signal.

	Elmrow, in this repo	Signal
First letter	Refuses to send unless a one-time device key exists. The letter key is the account key and that one-time key mixed together. Each is P-256 plus ML-KEM-1024. The text is AES-GCM-256. Up to four devices each get their own copy.	PQXDH: an X25519 identity key, a signed prekey, and a one-time prekey, mixed with ML-KEM-768.
Later letters	A new AES key for every message, from a hash chain. A new P-256 exchange and a new ML-KEM-1024 secret are mixed only on the first reply. After that, the chain just keeps going.	Every change of direction does a new X25519 ratchet. Since October 2025 the Triple Ratchet also keeps mixing a post-quantum secret.
Rooms and wards	One shared AES-GCM key for the room. Reactions use that same key.	A group protocol (Sender Keys, moving toward MLS) so one member is not a silent extra reader of a new epoch.
Calls	Normal browser WebRTC. Audio and video are DTLS-SRTP between the phones. Cloudflare only relays those packets. Setup (the offer, the answer, the network lines) is sealed with the letter keys. An extra frame lock exists and is switched off.	Media is encrypted inside Signal's own session, not left to WebRTC alone.
Private keys	On the device. The server stores ciphertext.	Same.
Outside review	No. This is Elmrow's own code.	The protocol is published and formally checked.

What that means

The server cannot read a letter. A stolen one-time key cannot open the next letter, because that key is deleted after it is used.

It does not keep healing the way Signal does. After the first reply, someone who stole the current chain key can derive the following message keys in that direction. Signal's ratchet throws those away on the next reply, including against a quantum attacker.

There is no safety number. A fingerprint is stored and a key change can raise an alert, but the prekeys are not signed and two people cannot compare a number the way they can in Signal.

The server still sees who wrote whom, and who is in a call. Signal hides the sender.

ML-KEM-1024 is a bigger post-quantum size than Signal's ML-KEM-768. That is the one place the letter seal is stricter. It does not make the whole session stronger than Signal's, because the ongoing ratchet stops mixing a new post-quantum secret after the first reply.

Private keys stay on the device. The server stores ciphertext. Calls are ordinary WebRTC: the browser encrypts the audio and video, and Cloudflare only forwards the packets. This note describes the code. It is not a security audit.